

End-to-End Customer Churn Analytics and Prediction System (Detailed Report)

1. Project Introduction

This project focuses on analyzing e-commerce customer data to understand behavior patterns and predict customer churn using machine learning techniques. Customer churn refers to customers who stop using a service or product. Predicting churn helps businesses take preventive actions and improve retention rates.

2. Problem Statement

Organizations face challenges in retaining customers due to low engagement, poor service quality, and increasing competition. Identifying customers who are likely to churn is critical for improving customer retention and maximizing revenue. This project addresses this problem using data-driven techniques.

3. Objectives

- Analyze customer behavior and identify patterns
- Understand factors influencing churn
- Build machine learning models for prediction
- Evaluate model performance
- Create dashboards for insights
- Deploy a Flask API for real-time prediction

4. Dataset Description

The dataset contains customer-related attributes such as age, membership duration, purchase behavior, login frequency, discount usage, and return rate. These features help in understanding customer engagement and predicting churn behavior.

5. Data Preprocessing

- Handled missing values using appropriate techniques
- Removed duplicate records
- Converted data types
- Normalized numerical features using StandardScaler

6. Exploratory Data Analysis (EDA)

EDA was performed to understand relationships between variables. Key analysis includes: • Age vs Churn • Purchase behavior vs Churn • Login frequency vs Churn • Correlation heatmap to identify feature relationships

7. Feature Engineering

Important features were selected based on business understanding and analysis: Age, Membership Years, Total Purchases, Average Order Value, Login Frequency, Days Since Last Purchase, Discount Usage Rate, Returns Rate.

8. Machine Learning Models

The following models were used: • Logistic Regression • Decision Tree • Random Forest Random Forest was selected as the final model due to better accuracy and ability to handle complex patterns.

9. Model Evaluation

Evaluation metrics used: • Accuracy • Confusion Matrix • Precision • Recall • F1 Score The final model achieved strong performance and generalization capability.

10. Key Business Insights

- Customers with low login frequency are more likely to churn
- Customers with fewer purchases show higher churn probability
- High return rate indicates dissatisfaction
- Long inactivity increases churn risk

11. Dashboard (Power BI)

An interactive dashboard was created with: • Total Customers KPI • Churn Rate KPI • Average Order Value • Customer segmentation • Filters and slicers for dynamic analysis

12. Flask API Deployment

A Flask API was developed to provide real-time predictions. Input: Customer data Output: Prediction + Probability This allows businesses to integrate the model into real applications.

13. Real-Time Workflow

1. Input customer data 2. Preprocess data using scaler 3. Predict churn using model 4. Return prediction and probability 5. Business team takes action

14. Business Applications

- E-commerce customer retention
- Banking churn prediction
- Telecom user retention
- OTT subscription analysis
- SaaS customer lifecycle management

15. Business Benefits

- Reduce customer loss • Increase revenue • Improve customer satisfaction • Better targeting strategies

16. Conclusion

This project demonstrates a complete end-to-end data science pipeline from data preprocessing to deployment. It showcases practical implementation of machine learning, analytics, and API integration, making it highly relevant for real-world business applications.